Basics of AI/ML/DL

Introduction

1. **Given a NumPy array representing a dataset, perform the following tasks:**
    1.1. Calculate basic statistics (mean, median, standard deviation) for each feature.
    1.2. Normalize the dataset so that each feature has a mean of 0 and a standard deviation of 1.
    1.3. Reshape the array from 2D to 3D (or vice versa)
2. **Python Functions and Loops:**
    2.1. Write a Python function that takes a list of numbers as input and returns a list with all elements squared.
    2.2. Create a loop that applies this function to each column in a NumPy array and stores the results in a new array.
3. **Loading and Inspecting Data:**
    3.1. Load the Iris dataset using Pandas and display the first few rows of the dataset.
    3.2. Display basic statistics for each feature (mean, max, min, etc.)
4. **Data Visualization:**
    4.1. Use Matplotlib to create scatter plots showing the relationships between pairs of features (e.g., sepal length vs. sepal width). Color the points based on the species.
    4.2. Create a histogram for each feature to visualize the distribution of values.
5. **Simple Analysis:**
    5.1. Calculate the correlation between features using Pandas and interpret the results.
    5.2. Filter the dataset to show only the records where the petal length is greater than 4 cm.
    5.3. Count the number of records for each species in this filtered dataset.
6. **Data Generation:**
    6.1. Create a simple dataset with one feature (e.g., house size) and a target variable (e.g., price) using NumPy.
7. **Model Implementation:**
    7.1. Implement the linear regression model from scratch (without using scikit-learn). The model should calculate the slope and intercept using the least squares method.
    7.2. Write a Python function that takes the dataset as input and returns the slope, intercept, and predicted values.
8. **Model Evaluation:**
    8.1. Calculate the mean squared error (MSE) between the predicted and actual values.
    8.2. Plot the regression line along with the data points using Matplotlib.